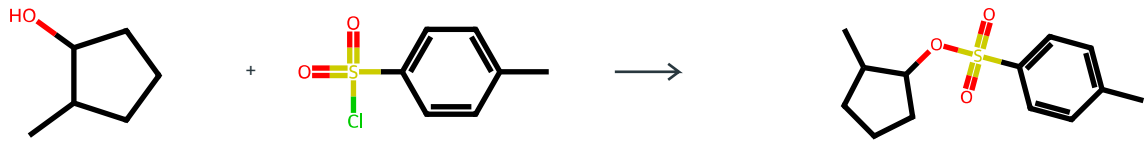
Performed experimental precedent for generated complete reactions

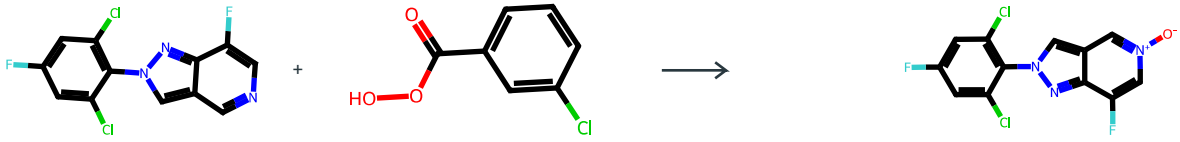
Selected cases; complete net chemistry, not experimentally established electron-arrow sequences.

1. Tier A · US5639915:reference8:2-methylcyclopentyl-tosylate



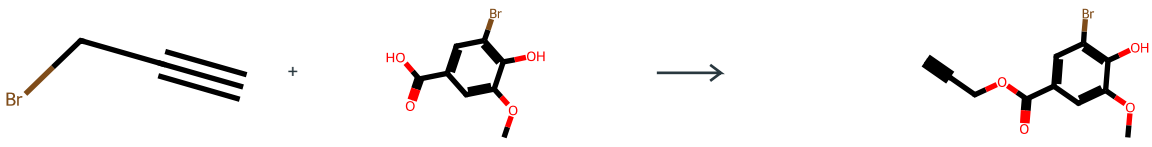
Alcohol 0.210 mol, tosyl chloride 48.3 g/0.252 mol, pyridine 170 mL added under ice cooling; stir 10 h while warming to room temperature. Ice-water/ether workup and acid/brine washes. 92.8%, clear oil, NMR; source mass OCR is retained in the witness, not silently repaired.

2. Tier A · US9309240B2:example43-step3:pyrazolopyridine-N-oxide



Heteroarene 3.37 g/11.2 mmol in DCM 50 mL under nitrogen; mCPBA 2.9 g/16.8 mmol, 0 C 2 h then room temperature 5 h. Thiosulfate/bicarbonate/brine workup, ether trituration; off-white solid 2.98 g, 84%, LCMS 316 (MH+).

3. Tier A · US4138579:example7:bromovanillic-propargyl-ester



5-Bromovanillic acid plus propargyl bromide, following Example 5 NaH/THF and hexamethylphosphoric triamide, reflux 2 h, aqueous/ether workup and chromatography. Product mp 121-122 C; no yield or substrate-specific amounts stated. Example 5 hydroquinone stoichiometry is not transferred.

Source/training overlap unknown; an experimentally documented transformation is not a tested whole route.